

Design and Implementation of an FPGA-based MIPI DSI Interface for Micro-LED Displays

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Abstract—This paper presents the design and implementation of an FPGA-based MIPI DSI interface optimized for driving Micro-LED displays. The proposed design addresses key limitations of the Xilinx MIPI DSI Subsystem, particularly the lack of support for Low-Power (Escape Mode) commands, non-continuous clock modes, and limited flexibility in lane configuration. The FPGA-based solution supports dynamic lane speed adjustment and offers flexible lane configurations (1, 2, or 4 lanes), ensuring High-Speed data transmission compatible with various Micro-LED display systems. Key innovations include robust clock management and scalability, enabling efficient operation in both Low-Power and High-Speed modes. Simulation and hardware testing confirm that the design achieves lane rates of up to 1.5 Gbps, ensuring reliable data transfer for high-resolution displays. The results demonstrate significant improvements in performance, flexibility, and synchronization compared to existing solutions.

Index Terms—MIPI DSI, FPGA-based design, Micro-LED displays, dynamic lane speed adjustment, flexible lane configuration

I. INTRODUCTION

MIPI DSI (Display Serial Interface) has emerged as a critical standard for High-Speed data transmission in modern display systems [7], particularly for Micro-LED technology. Micro-LED is an advanced display technology known for its superior resolution, brightness, and energy efficiency [8]. It requires High-Speed data transmission and low power consumption [5], which presents significant challenges for existing MIPI DSI solutions. While MIPI DSI is well-suited for delivering high-resolution displays with low power consumption, the existing Xilinx MIPI DSI Subsystem exhibits several limitations that hinder its performance and flexibility, particularly in the context of advanced applications such as Micro-LED displays.

Specifically, the current Xilinx MIPI DSI Subsystem faces the following challenges [4]:

- 1) It does not support the necessary commands for properly initializing Micro-LED displays during startup.
- 2) It does not support Low-power (LP) mode data transmission (Escape Mode), limiting its functionality in Low-Power scenarios.
- 3) It lacks support for non-continuous clock modes, which are critical for certain display applications.
- 4) It cannot handle flexible lane configurations, which are essential for adapting to varying display system requirements.

To address these challenges, we propose an FPGA-based MIPI DSI interface that overcomes these limitations and is optimized for driving Micro-LED displays. Our design supports flexible lane configurations, including options for 1, 2, or 4 lanes, enabling it to adapt to various display system requirements. A key feature of the design is its ability to dynamically adjust the speed of the lane, providing improved flexibility and performance compared to existing solutions. This capability ensures seamless data transmission in both Low-Power and High-Speed modes.

Additionally, the design resolves clocking challenges by supporting both continuous and non-continuous clock modes. This flexibility is essential for Micro-LED displays, which require adaptable power consumption and data transfer speeds. Robust clock management ensures reliable data synchronization, even with varying resolution configurations.

Our FPGA-based interface also addresses synchronization issues between the master and slave devices, ensuring real-time transmission of accurate image data to the Micro-LED displays. This enhancement results in smooth operation without clock drift.

Through extensive simulation and hardware testing, we demonstrate that our design achieves a maximum lane rate of 1.5 Gbps, ensuring High-Speed and reliable data transfer. These results confirm that the proposed FPGA-based MIPI DSI interface significantly improves upon the Xilinx MIPI DSI Subsystem in terms of protocol support, lane configuration flexibility, and clock management.

This paper provides an effective and scalable solution for FPGA-based display systems, offering improved flexibility and reliability for power-efficient, high-resolution display applications.

II. SYSTEM OVERVIEW

The system is designed with a clear Master-Slave architecture, where the Master manages the data flow, and the Slave (Micro-LED display) receives and processes the data for display.

As shown in Fig. 1., the Master handles the image or video data sourced from SD and DDR memory. The initialization commands are stored in the Config ROM and passed to the Initialization Commands module for proper system setup. The image data is buffered in the Dual-Clock FIFO, which synchronizes the clocks needed for image processing and

transmission. After buffering, the data are converted into RGB888 format using the RGB888 conversion module for MIPI DSI transmission.

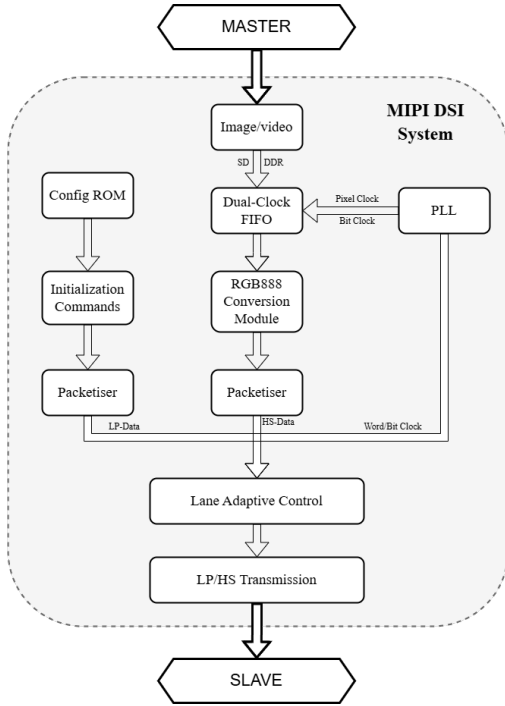


Fig. 1. System framework diagram.

The data is then processed by two packetisers: one handles LP data, and the other handles HS (High-Speed) data, ensuring compatibility with the MIPI DSI protocol. The Lane Adaptive Control module adjusts the transmission based on the lane count and manages transitions between LP and HS modes to optimize power efficiency and data transfer speed.

Once processed, the data are sent through the LP/HS Transmission module, which controls switching between Low-Power and High-Speed modes as needed. Finally, the Slave module, the Micro-LED display, receives and displays the data. The system's timing is managed by a PLL, which generates the necessary Pixel and Bit clocks for synchronization.

This architecture ensures efficient data flow from the Master to the Micro-LED display, supporting flexible and high-performance display operation.

III. DESIGN AND IMPLEMENTATION

A. Initialization Module

One of the key innovations of this design is the inclusion of an initialization module that supports the Low-Power (Escape Mode) command. The Initialization Module is designed to automatically initialize the Micro-LED display upon power-up by sending specific commands stored in the Config ROM. These commands include essential instructions like the **DCS Long Write** command (with a Data Type of 0x39) that prepares the display for proper operation. After powering on, the system retrieves and transmits these commands to the

display, ensuring it is configured correctly before it starts displaying image data.

B. HS/LP Mode Conversion Module

The HS/LP Mode Conversion Module addresses the limitation of the existing system, which does not support LP mode data transmission (Escape Mode). It effectively manages the transition between Low-Power (LP) and High-Speed (HS) modes in the MIPI DSI protocol. By switching between Low-Power operation and High-Speed data transmission, this module ensures efficient power usage and high data throughput, which is critical for applications like Micro-LED displays that require both power efficiency and high performance.

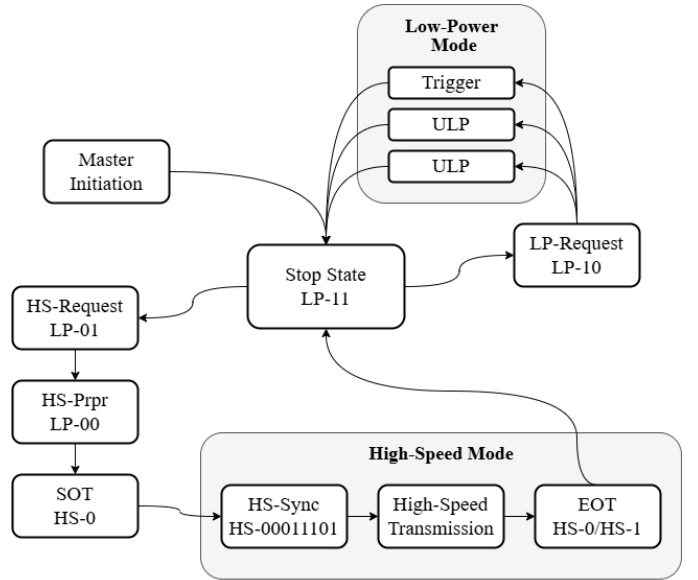


Fig. 2. HS/LP mode conversion state machine.

As illustrated in Fig. 2, the process begins with Master Initialization. Afterward, the system moves into Low-Power Mode, entering the Stop State (LP-11) to cease active transmission. An LP-Request (LP-10) signals the transition, putting the system into a reduced power state [2], [3]. Within LP mode, the system can enter two sub-states: Trigger and Ultra-Low Power (ULP), optimizing power consumption until High-Speed data transmission is needed again [1]. This transition ensures the system efficiently switches between Low-Power and High-Speed modes as required by the application.

Next, in High-Speed Mode (HS), the system receives the HS-Request (LP-01) and HS-Prpr (LP-00) signals to transition into High-Speed data transmission. The system then sends the Start of Transmission (SOT) signal (HS-0) to initiate data transfer. Synchronization is achieved through the HS-Sync phase, and once synchronization is complete, High-Speed data transmission begins. Finally, upon completion of data transmission, the system sends the End of Transmission (EOT) signal (HS-0/HS-1), marking the end of the High-Speed data transfer.

C. MIPI DSI Transmitter Module

The MIPI DSI transmitter module is responsible for converting parallel data into the MIPI DSI format, supporting both Low-Power (LP) and High-Speed (HS) data transmission modes. In LP mode, it optimizes power consumption by using asynchronous data transmission with a large swing signal (e.g., 1.2 V) for efficient communication. In HS mode, it ensures high-speed data transfer with synchronous NRZ signals based on scalable low voltage differential signaling (SLVS), utilizing a common-mode voltage of 0.2 V. This enables efficient and reliable communication in Micro-LED display applications [6].

1) *Low-Power Mode Transmission:* Low-Power (LP) Mode reduces power consumption by minimizing data transmission, using timing derived from the Dp and Dn lines via XOR. Escape Mode, a form of LP Mode, allows asynchronous communication, as shown in Fig. 3 [3]. The transition starts with LP-11, progresses through LP-10 and LP-00, and enters Escape Mode in the Space state. If LP-11 is detected early, the procedure is aborted and returns to the Stop state.

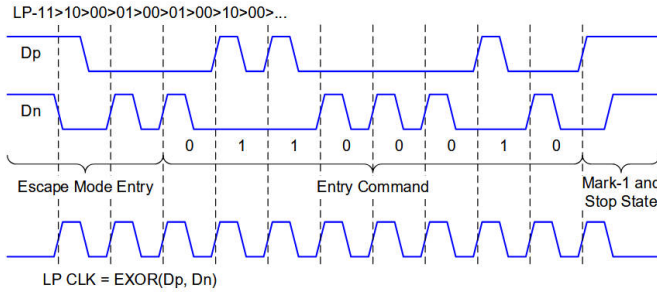


Fig. 3. Trigger-Reset command in escape mode.

2) *High-Speed Mode Transmission:* In High-Speed (HS) mode, data is transmitted in bursts. The system transitions from LP-11 to LP-01 to LP-00 before entering HS mode [2], [3]. Data transmission starts after a synchronization period, and the system allows minor errors in the start sequence. Each burst is preceded by a leader and followed by a trailer sequence for synchronization. After the burst, the system enters the End-of-Transmission (EoT) procedure and returns to Stop state, either initiating a new transmission or going back to LP mode. This ensures High-Speed data flow, as defined by the MIPI DSI protocol.

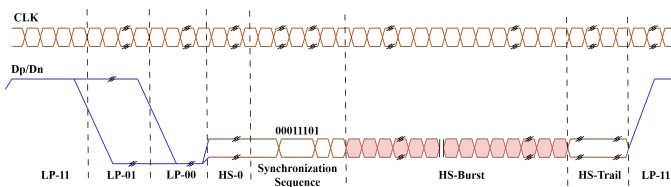


Fig. 4. High-Speed data transmission in bursts.

D. Clock Management Module

This module directly addresses the limitation in the existing system that lacks support for non-continuous clock modes, a key requirement for certain display applications. The Clock Management Module generates and controls the clocks needed for MIPI DSI transmission, ensuring that the system can operate efficiently under both continuous and non-continuous clock modes.

In Low-Power mode, the system disables the clock and transmits data using Spaced-One-Hot encoding [3]. This approach allows data to be transmitted asynchronously without relying on a continuous clock signal, as shown in Fig. 3. The Slave device, which receives the data, can extract the clock signal from the transmitted data through an XOR operation between the Dp and Dn lines. This method enables reliable data transmission in LP mode while minimizing power consumption.

When transitioning to High-Speed (HS) mode, the system enables the clock, providing the necessary synchronization for High-Speed data transfer. The Clock Management Module dynamically adjusts the clock frequency based on the current data transmission mode, ensuring that both LP and HS modes operate effectively. This flexibility is essential for display systems such as Micro-LED, where both power efficiency and High-Speed data transmission must be balanced. The diagram in Fig. 5 illustrates the clock characteristics in both LP and HS modes. It demonstrates the power-efficient clock management during LP mode and the High-Speed data transfer during HS mode, showcasing the adaptability of the Clock Management Module across varying display requirements.

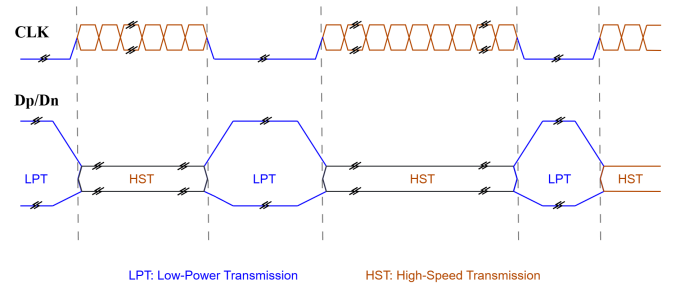


Fig. 5. Clock level characteristics in LP and HS modes.

E. Flexible Lane Configuration Adapter Module

The Flexible Lane Configuration Adapter Module solves the challenge of handling variable lane configurations, a critical requirement for adapting to diverse display system needs. It ensures adaptability for a wide range of display systems.

The module operates by splitting the 32-bit input data into two 8-bit segments, each assigned to a separate lane. It also manages the control signals, such as Enable and End-of-Transmission (EoT), to ensure proper synchronization of data across the two lanes. The system detects the start and end of the data packet, maintaining smooth transmission and correct packet handling.

This solution provides the flexibility to switch between different lane configurations (1, 2, or 4 lanes), making it highly adaptable to various display requirements. It supports high-resolution displays, offering an efficient method for adapting lane configurations according to specific needs.

IV. EXPERIMENTAL RESULT

A. Simulation Results

The simulation, conducted using Vivado, verifies the system's data flow from the host to the Micro-LED display, as shown in Fig. 6. Initially, image data is stored in the FIFO based on VGA timing. Once the system enters the valid display area, the host writes the data into the FIFO. Upon transitioning to High-Speed (HS) mode, the data is read, packed into long packets, and transmitted bit-by-bit through two data lanes to the Micro-LED display, where the grid image appears. This simulation confirms the correct formatting and transmission of data in both LP and HS modes.

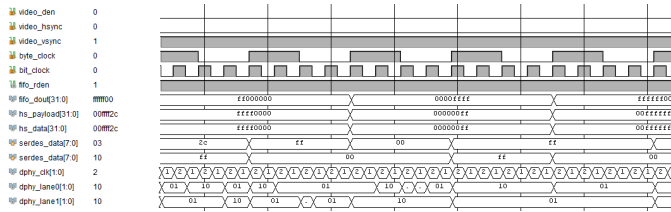


Fig. 6. Simulation waveform of data transmission to Micro-LED display.

B. Hardware Verification

To verify the hardware implementation, we connected the ARTIX UltraScale+ FPGA (Model: AXAU10) to the Micro-LED display and observed the results under a microscope. The system successfully transmitted the image data to the Micro-LED, which displayed the grid pattern as expected. As shown in Fig. 7, the FPGA was interfaced with the Micro-LED display, and the grid pattern was visible on the display.

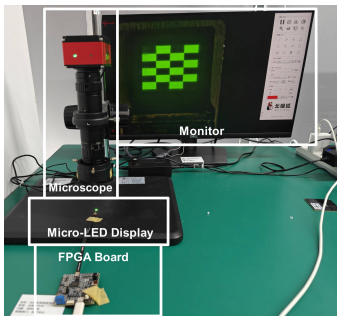


Fig. 7. FPGA and Micro-LED under the microscope.

The microscope image in Fig. 8 further confirms the accurate display of the grid pattern on the Micro-LED. This image demonstrates that the FPGA correctly processes and sends the data, which is displayed clearly and without distortion on

the Micro-LED screen. This hardware verification ensures the system's functionality and reliability in real-world conditions.

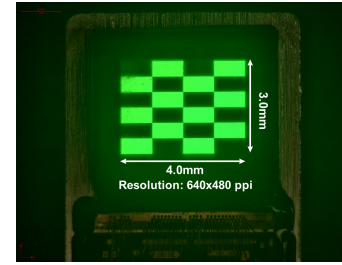


Fig. 8. MIPI DSI interface drives the Micro-LED display image.

V. CONCLUSION

In this paper, we proposed an FPGA-based MIPI DSI interface designed specifically for driving Micro-LED displays. The design addresses key limitations in the existing Xilinx MIPI DSI Subsystem, including the lack of support for Low-Power (Escape Mode) commands, non-continuous clock modes, and flexible lane configurations. The system offers dynamic lane speed adjustment and robust clock management, enabling seamless data transmission in both Low-Power and High-Speed modes. Experimental results show that the system can achieve a maximum lane rate of 1.5 Gbps, ensuring High-Speed and reliable data transfer to Micro-LED displays. This FPGA-based design provides a flexible and scalable solution, improving performance and adaptability for high-resolution, power-efficient display applications.

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